

Coverage Report

Project 7 — NFT Event Ticketing (ERC-721)

solidity-coverage v0.8.17 / Istanbul

Environment Configuration

Command: `npm run coverage (npx hardhat coverage)`

Solc: 0.8.28

Optimizer: enabled

Runs: 200

Network: hardhat (chainId 31337)

HardhatEVM: v2.28.6

Instrumenting for coverage...
contracts/EventTicketNFT.sol

Compilation

Nothing to compile.
No need to generate any newer typings.

Test Results: EventTicketNFT

Deployment

- ✓ deploys with correct name and symbol
- ✓ sets deployer as owner
- ✓ starts with zero events and zero tokens
- ✓ supports ERC-721 and ERC-2981 interfaces

createEvent

- ✓ stores event with a single section and emits EventCreated
- ✓ stores multiple sections and sets priceWei to the cheapest
- ✓ emits SectionCreated for each section
- ✓ reverts with empty metadataURI
- ✓ reverts if date is in the past
- ✓ reverts if no sections are provided
- ✓ reverts if a section has maxTickets = 0
- ✓ reverts if royalty exceeds cap (50 %)
- ✓ reverts if maxPerBuyer is zero
- ✓ reverts if maxPerBuyer exceeds GLOBAL_MAX_PER_BUYER
- ✓ increments eventId counter

buyTicket

- ✓ mints NFT from chosen section and transfers ETH to organiser
- ✓ charges each section its own price
- ✓ records per-token section
- ✓ sets tokenURI derived from event metadata + tokenId
- ✓ refunds excess ETH to buyer
- ✓ reverts if event does not exist
- ✓ reverts with invalid section id
- ✓ reverts with insufficient payment
- ✓ reverts when a section is sold out
- ✓ reverts when event date has passed
- ✓ reverts when event is cancelled
- ✓ enforces per-buyer cap across sections

buyMultipleTickets

- ✓ mints multiple NFTs from the same section
- ✓ reverts if section supply exceeded
- ✓ reverts if per-buyer cap exceeded
- ✓ reverts with quantity 0
- ✓ reverts with insufficient payment

listForResale

- ✓ lists ticket owned by caller and emits event
- ✓ adds tokenId to active listings
- ✓ reverts if caller is not owner
- ✓ reverts if already listed
- ✓ reverts if price is zero
- ✓ reverts when listing an invalidated ticket
- ✓ rejects expiry in the past
- ✓ rejects expiry after event date
- ✓ allows listing exactly at 2x the original price
- ✓ reverts if resale price exceeds 2x the original
- ✓ maxResalePriceFor returns 2x original price

buyResaleTicket

- ✓ splits payment 90/10 (royalty 10 %) and transfers ownership
- ✓ reverts if seller tries to buy their own listing
- ✓ reverts when event is cancelled
- ✓ reverts when listing was cancelled by the seller
- ✓ reverts when listing was auto-cancelled by ticket invalidation

royaltyInfo (EIP-2981)

- ✓ returns (organiser, 10 % of salePrice)

Organiser admin

- ✓ addTicketsToSection increases supply
- ✓ addTicketsToSection reverts when called by non-organiser
- ✓ addTicketsToSection reverts with amount 0
- ✓ addTicketsToSection reverts with invalid section id
- ✓ cancelEvent by organiser blocks further sales
- ✓ invalidateTicket by organiser cancels any listing

updateEvent

- ✓ organiser can change mutable fields before first sale
- ✓ royalty is locked once a ticket is sold
- ✓ rejects empty metadataURI
- ✓ reverts for non-organiser

View helpers

- ✓ getTicketsOfUser returns correct ids
- ✓ ticketsBoughtBy reflects per-buyer counter
- ✓ getEventOfToken returns event data for a token
- ✓ getSectionOfToken returns the section the ticket belongs to
- ✓ getActiveListings reflects additions and removals

61 passing (3s)

Coverage Summary

Reports written to ./coverage/ and ./coverage.json

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Lines
contracts/	97.21	72.50	100.00	99.10	
EventTicketNFT.sol	97.21	72.50	100.00	99.10	629, 630
All files	97.21	72.50	100.00	99.10	

Rubric Check §E.3

“Coverage report showing ≥ 70 % line coverage” (1 mark)

Statements : 97.21 % (target ≥ 70 %) → **PASS**

Branches : 72.50 % (target ≥ 70 %) → **PASS**

Functions : 100.00 % (every public/external function exercised)

Lines : 99.10 % (target ≥ 70 %) → **PASS**

Uncovered Lines (629–630)

The defensive excess-ETH refund branch in `buyResaleTicket`. It only fires when the buyer over-pays beyond `listing.price`. The tests always send `{ value: salePrice }` exactly, so the `excess > 0` branch is never taken. The equivalent refund branch in `buyTicket` IS covered by the test “refunds excess ETH to buyer”.

Reproduce

```
npm install
npm run coverage
xdg-open coverage/index.html      # full per-line HTML report
```